

# Modelling how children with Down syndrome learn language

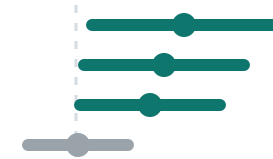
Two studies: how vocabulary grows, and what reading-and-language teaching causes.



## STUDY 01

### Vocabulary growth

How children understand, say and sign words as they grow, across eleven international studies.



## STUDY 02

### Reading & language

A re-analysis of a randomised trial, separating what the teaching causes from what tracks ability.

# One method, two studies

Both studies run on the same Bayesian foundation. What changes is the question, and how far the data can be pushed toward cause.

STUDY 01 · VOCABULARY GROWTH

## Vocabulary growth

*How do children build words?*

Models the words children **understand, say and sign together**, pooled across eleven international studies. The comprehension–production gap falls straight out of the model as the production ratio **q**.

WHAT’S DISTINCTIVE

The understood × production-ratio decomposition, estimated jointly across eleven studies.

STUDY 02 · READING & LANGUAGE

## Reading & language

*What does teaching cause?*

Re-analyses a **randomised trial** through a locked causal DAG, so the teaching effect is identified by randomisation. Gradient-boosting discovery feeds the Bayesian estimates.

WHAT’S DISTINCTIVE

A causal DAG plus randomisation; a two-layer discovery-then-estimation method.

THE SHARED FOUNDATION

Hierarchical Bayesian models

Beta–Binomial likelihood

Gaussian–process (HSGP) trends

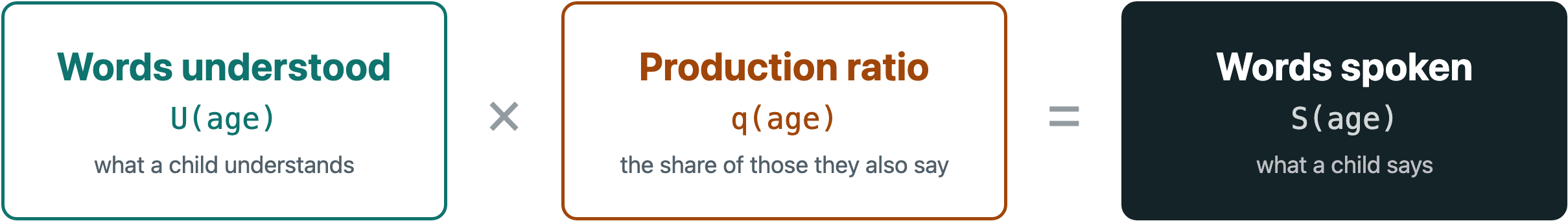
credible intervals throughout

reproducible: PyMC · NUTS · DuckDB · Quarto

The same engine underneath: **full-posterior inference**, checked with predictive checks and cross-validation, and reproducible from raw data to report.

# How the vocabulary model works

The modelling idea at the centre of the study: spoken vocabulary is built from what a child understands and the share of it they say.



Modelling the two together means spoken vocabulary can **never exceed** what a child understands, and the comprehension-production gap falls out directly as the production ratio **q**.

## THE DATA

Parent-report vocabulary checklists, **pooled across 11 international studies** onto a common 810-word scale: **576 children**, ages 8 months to about 9½ years, roughly 1,000 observations.

## THE ESTIMATE



Each term is a **smooth curve over age** with full posterior uncertainty, so every result is a credible range, not a single number.

Hierarchical Beta-Binomial

Gaussian-process (HSGP) trends

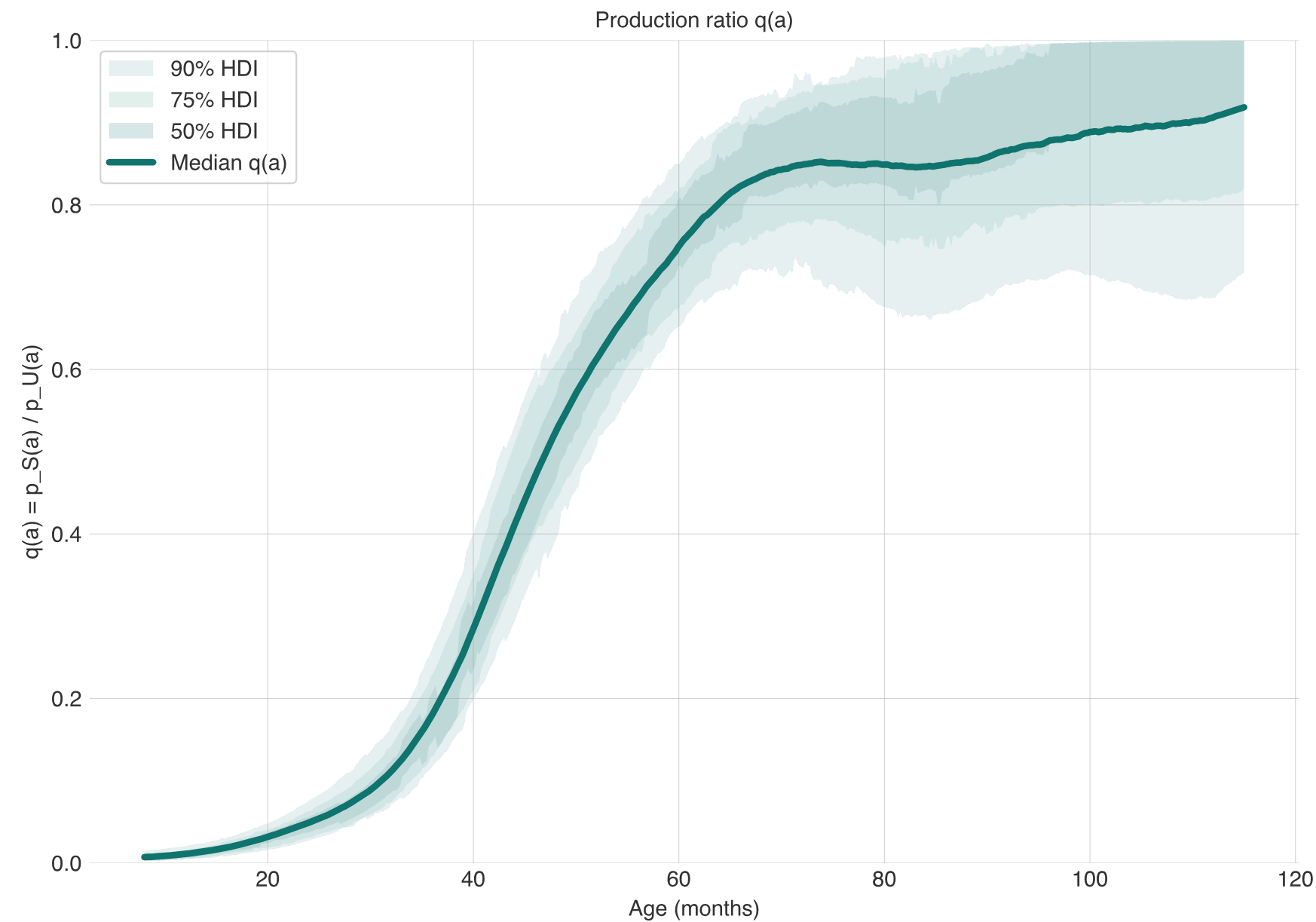
study & child random effects

PyMC / NUTS

PSIS-L00 comparison

# Understanding runs years ahead of speaking

The share of understood words a child with Down syndrome also says, estimated across age from pooled international data.



Each child's **production ratio** is the fraction of the words they understand that they also say. Comprehension leads production for years, so the ratio stays low at first, then climbs steeply.

~1%

at 1 year

~52%

at 4 years

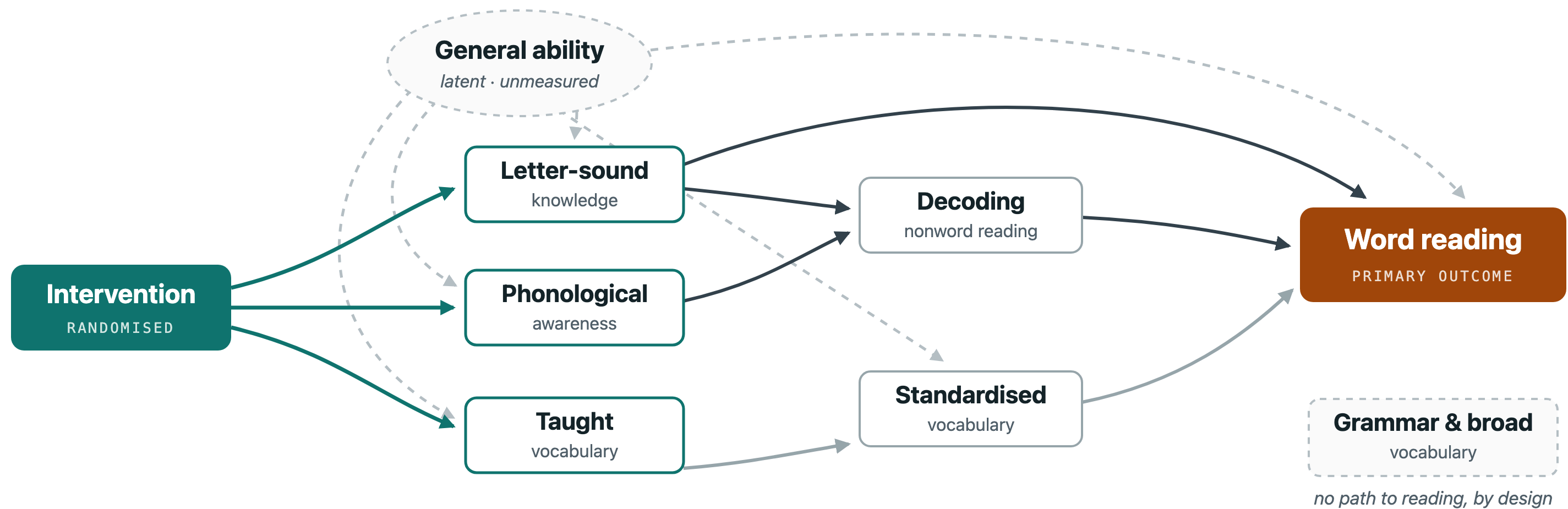
~86%

by 7½ years

**Wider than in typically-developing children** at low vocabulary levels, narrowing as vocabulary grows.

# A causal design for reading progress

Separating what the teaching *causes* from what merely correlates with a child’s general ability.



## HOW TO READ IT

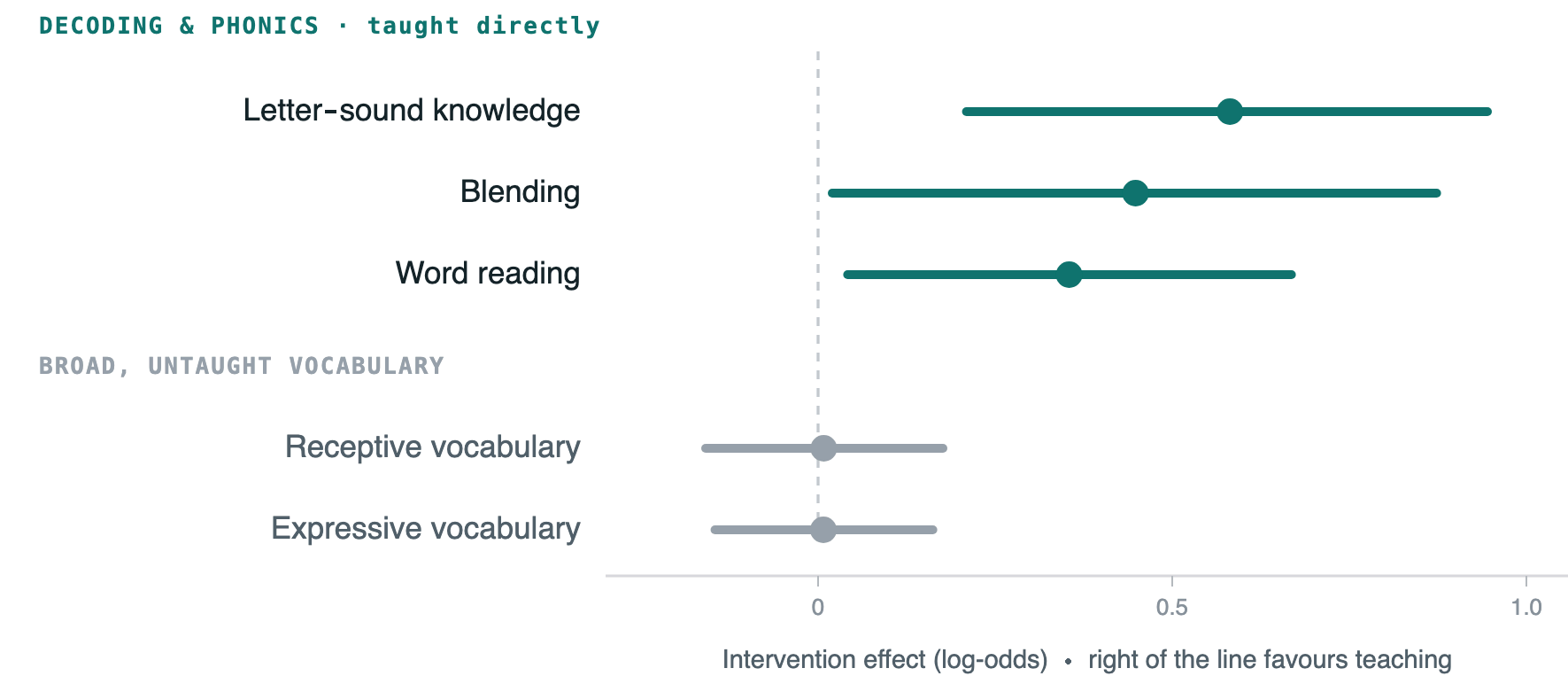
- Randomised effect, **causal**
- Hypothesised skill pathway, association
- General-ability confounding
- Association through standardised vocabulary
- Teal outline: a skill the teaching targets directly

## THE IDENTIFICATION

**Randomisation** makes the intervention independent of general ability, so its effect on word reading is read as **genuine cause**. Every skill-to-skill link stays confounded by ability, so it is reported as **association, not cause**.

# The intervention lifted reading, not broad vocabulary

The randomised effect of the teaching on each skill.



Each bar is the intervention's effect on one skill, with its **95% credible interval**. Children were randomised, so these read as cause, not correlation.

The gains cluster on the skills taught most directly: letter-sounds, blending and word reading. Broad, untaught vocabulary barely moved.

**Learning transfers near what's taught, not far.** The effect fades with distance from instruction. This is the transfer gradient.

● A bar clear of the dashed zero line is a credibly non-zero effect.